SUDHANSHU KUMAR

Senior Cloud Engineer | Azure & AWS | Kubernetes | IaC | DevOps

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PROFESSIONAL SUMMARY

Senior Cloud Engineer with 5.5+ years of hands-on experience architecting and automating enterprise-scale cloud infrastructure, primarily on Microsoft Azure with hands-on exposure to AWS for multi-cloud and migration initiatives. Specialized in AKS/EKS, Terraform IaC, CI/CD pipelines, and full-stack observability. Proven track record of reducing cloud costs by 20%, cutting incident response times by 50%, and delivering zero-downtime deployments under peak load. AZ-204 certified. Currently owning end-to-end Azure infrastructure design and delivery at R Systems International.

TECHNICAL SKILLS

Cloud Platforms: Microsoft Azure (AKS, ACR, App Gateway, WAF, Front Door, Key Vault, VNet, NSG, Azure Policy) — primary; AWS (S3, IAM, Glue, Lambda, Step Functions, API Gateway, EventBridge, SNS, VPC, Secrets Manager, RDS) — secondary/multi-cloud project experience

Infrastructure as Code: Terraform (v0.13 to v1.x, Terraform Cloud/HCP), Azure Bicep, ARM Templates

Containers & Orchestration: Kubernetes (AKS, EKS), Docker, Helm, Container Registry

CI/CD & DevOps: Azure DevOps, GitHub Actions, Jenkins, Git

Monitoring & Observability: Azure Monitor, Log Analytics, Application Insights, Prometheus, Grafana, AWS CloudWatch

Scripting & Automation: Python, Bash, PowerShell

Security & Governance: RBAC, Azure Policy (custom governance initiatives), IAM, CIS Benchmarks, Key Vault, WAF, Azure Front Door Premium, Private Link, TFLint, Checkov

Familiar With: Ansible, MySQL, Android, Java, C++

PROFESSIONAL EXPERIENCE

Senior Cloud Engineer, R Systems International *Dec 2024 – Present | New Delhi, India*

Key Project — PeriGen (Vigilance Mobile): Azure Cloud Platform

- Architected end-to-end Azure infrastructure using **Terraform**, delivering **17+ reusable modules** for networking, AKS, security, and observability across dev/staging/prod.
- Built a private **AKS cluster** (K8s 1.30) with Workload Identity, OIDC issuer, Azure CNI overlay networking, and cluster autoscaling, on a hub-and-spoke network topology with segmented subnets and NSGs.
- Implemented **Azure Front Door Premium + Private Link + WAF** for secure, zero-public-exposure ingress to Kubernetes workloads (AKS behind internal NGINX load balancer).
- Established Azure Policy governance with **10+ custom audit policies** for security, cost control, tagging, and compliance, plus Key Vault (RBAC) for centralized secrets management.
- Automated infrastructure delivery via **Azure DevOps pipelines** with staged Terraform plan/apply, gated approvals, and integrated **TFLint/Checkov** security scanning.
- Built AKS monitoring with Container Insights, kube-audit logging, and metric alerts (CPU, memory, disk pressure, CrashLoopBackOff) routed to email action groups via dual Log Analytics workspaces.

Additional Project — AWS Serverless Data Platform (GHG Emissions Calculation)

- Designed and deployed production AWS infrastructure via **Terraform** (S3, IAM, Glue, Lambda, Step Functions, SNS, API Gateway) across **4 environments**, using modular reusable modules and Terraform Cloud remote state.
- Architected an event-driven **serverless pipeline** with AWS Step Functions to orchestrate multi-stage ETL workflows, using Glue 5.0 jobs with secure RDS/JDBC connectivity and Lambda-backed API Gateway endpoints.
- Built **GitHub Actions CI/CD** pipelines for Terraform with AWS OIDC (no long-lived credentials), automating Glue connection updates and path-triggered ETL job deployments.
- Configured VPC-attached Lambda functions with least-privilege IAM policies and Secrets Manager integration; implemented SNS alerting and EventBridge-scheduled orchestration runs for operational monitoring.

Software Developer (Cloud & DevOps), Tata Consultancy Services *Jan 2022 – Dec 2024 | New Delhi, India*

- Led Angular migration of a critical client application and Gen2+ Databricks migration using VNet injection, improving platform performance by **30%** and reducing compute costs by **20%**.

- Engineered multi-stage Azure DevOps pipelines with Terraform, reducing production deployment cycle time by **40%** and enabling repeatable, auditable infrastructure delivery across **5+ client environments**.
- Drove a **75%** improvement in application stability through structured load testing and auto-scaling configuration, achieving a zero-downtime SLA during peak traffic for a production workload.
- Built an automated monitoring system for Azure Function Apps with real-time failure alerting via email, reducing mean incident response time by **50%** and enabling proactive issue resolution.
- Upgraded Terraform from v0.13 to v1.2, migrating deprecated resources and aligning configurations with current provider standards — eliminating technical debt across **12+ modules**.
- Executed Inox@scale and GitHub migrations, consolidating repositories and reducing licensing and tooling costs by **20%**.
- Developed Python-based automation scripts to parse and aggregate Azure Log Analytics data, generating scheduled health reports and dashboard exports — reducing manual reporting effort by **70%**.
- Built a Python utility to automate Azure Resource Manager (ARM) template validation and drift detection across environments, catching configuration mismatches before deployments and preventing **10+ production incidents**.
- Developed RESTful backend microservices in Python (Flask/FastAPI) to expose Azure infrastructure metrics and deployment status via internal APIs, enabling self-service dashboards consumed by **3 development teams**.
- Built Java-based backend services integrated with Azure Event Hubs and Service Bus to process real-time data streams, supporting event-driven architecture for client applications handling **50,000+ daily transactions**.

Assistant System Engineer, Tata Consultancy Services *Jan 2021 – Jan 2022 | Pune, India*

- Designed and implemented a long-term data retention and backup strategy using Terraform, reducing data loss risk by **90%** and ensuring compliance with client SLAs.
- Refactored Azure Pipelines to introduce distinct development and production stages, improving deployment efficiency by **25%** and reducing environment misconfiguration incidents.
- Led an Agile cross-functional team of 5 engineers, improving project delivery velocity by **15%** through structured sprint planning and client requirement alignment.
- Delivered internal training on Azure services (Function Apps, Application Insights, Key Vault) and DevOps tooling (GitHub, Docker, Kubernetes, Ansible), increasing team onboarding speed by **20%**.

CERTIFICATIONS

- AZ-204: Microsoft Azure Developer Associate — Microsoft Certified

EDUCATION

B.Tech in Computer Science — GGSIPU, Delhi *Graduated Aug 2020*